

RESEARCH ARTICLE

Weakening Ion Transport Anisotropy Activates Superionic Conduction in Li_2ZrCl_6

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ABSTRACT

Halide solid-state electrolytes (SSEs) are promising for all-solid-state batteries (ASSBs) but are often limited by the challenge of simultaneously achieving sufficient ionic conductivity and cost-effectiveness. Herein, we demonstrate that weakening ion transport anisotropy is an effective route to achieve superionic conduction in low-cost Li_2ZrCl_6 through an anion-cation co-doping strategy using InF_3 . Mechanistic analysis reveals that moderate co-doping reduces the intrinsic anisotropy of Li-ion transport, thereby facilitating faster long-range diffusion, whereas excessive doping introduces localized migration bottlenecks in the monoclinic lattice, leading to diminished ionic conductivity. The optimized $\text{Li}_{2.1}\text{Zr}_{0.9}\text{In}_{0.1}\text{Cl}_{5.7}\text{F}_{0.3}$ exhibits a high ionic conductivity of $1.03 \times 10^{-3} \text{ S cm}^{-1}$ with a low activation energy of 0.357 eV. As a result, the ASSBs using this optimized halide SSE deliver high reversible capacity, excellent rate capability, and long-term stability, retaining 92.5% capacity after 500 cycles at 1 C and operating stably over 2500 cycles at 4 C. Moreover, the enhanced ionic conductivity enables stable operation at a high cathode loading of 31.2 mg cm^{-2} . This work highlights weakening ionic transport anisotropy as an effective strategy for designing high-performance halide SSEs.

1 | Introduction

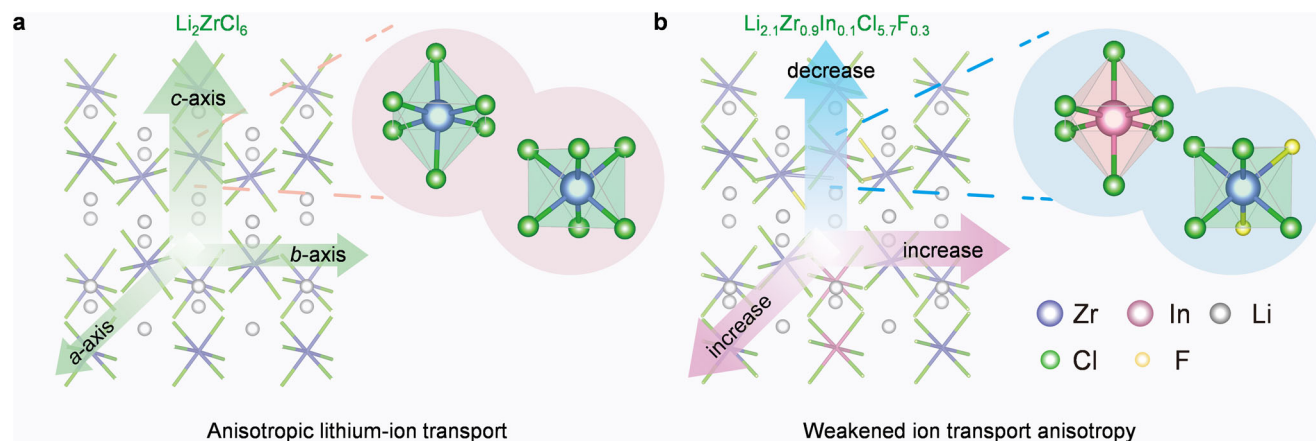
All-solid-state batteries (ASSBs) are widely regarded as next-generation energy storage systems due to their promising enhanced safety, high energy density, and long cycle life [1, 2]. As the critical component, solid-state electrolytes (SSEs) largely determine the overall battery performance by governing ionic transport, interfacial stability, and electrochemical com-

patibility with electrodes [3–7]. Among various SSE families, including oxides and sulfides [8, 9], halide-based electrolytes have recently emerged as highly promising candidates because of their balanced properties, such as high room-temperature ionic conductivity (10^{-4} – $10^{-3} \text{ S cm}^{-1}$), wide electrochemical stability windows, and friendly environmental compatibility [10–16]. Among halide-based SSEs, Li_2ZrCl_6 (LZC) stands out as a particularly attractive candidate for practical applications

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SCHEME 1 | (a) The anisotropic lithium-ion transport in LZC structure; (b) Weaken ion transport anisotropy through synergistic anion-cation doping in $\text{Li}_{2.1}\text{Zr}_{0.9}\text{In}_{0.1}\text{Cl}_{5.7}\text{F}_{0.3}$ structure.

owing to the use of earth-abundant elements, which offer distinct cost advantages over rare-earth-based halides [17–19]. However, the relatively low ionic conductivity of LZC SSEs still hinders practical applications.

To address the limitation and achieve a more well-balanced performance profile, various modification strategies have been proposed. Cation doping has been widely employed to tailor lattice structure, carrier concentration, and configurational entropy, thereby improving ionic conductivity [20–24]. In contrast, anion substitution has been primarily explored to regulate electrochemical stability, although its influence on ionic transport remains less understood and often system-dependent [25–28]. However, these single-element doping strategies typically optimize only one aspect of performance, making it difficult to simultaneously enhance both ionic conductivity and interfacial stability. Moreover, there is rare in-depth discussion on the performance evolution induced by cation-anion co-doping in the existing research. Studies on dual (multi-) element doping caused ionic conductivity and interfacial stability variation often remain at a mesoscopic level, despite the fact that the doping ions carry the homocharge, making it difficult to fundamentally elucidate and understand the underlying relationships among composition, structure, and properties in SSEs. Therefore, developing a synergistic dual-doping strategy that can concurrently regulate structural and interfacial properties is of great importance [29, 30].

Herein, we propose an anion-cation co-doping strategy using InF_3 to simultaneously enhance the ionic conductivity and electrochemical stability of LZC. The optimized $\text{Li}_{2.1}\text{Zr}_{0.9}\text{In}_{0.1}\text{Cl}_{5.7}\text{F}_{0.3}$ composition exhibits a high ionic conductivity of $1.03 \times 10^{-3} \text{ S cm}^{-1}$, nearly doubling that of pristine LZC. Combined experimental and theoretical investigations reveal that the synergistic incorporation of In^{3+} and F^- weakens the intrinsic anisotropy of Li^+ transport by redistributing diffusion pathways and thus enables more balanced lithium migration along each direction, as illustrated in Scheme 1. Meanwhile, the co-doping strategy improves interfacial stability against lithium metal through the formation of stable interphases. As a result, ASSBs based on the optimized electrolyte demonstrate excellent rate capability and long-term cycling stability. This work establishes transport

anisotropy engineering via dual-ion doping as an effective design principle for developing high-performance halide SSEs.

2 | Results and Discussion

The preparation of InF_3 -doped LZC ($\text{Li}_{2+x}\text{Zr}_{1-x}\text{In}_x\text{Cl}_{6-3x}\text{F}_{3x}$, denoted as LZICF) and pristine LZC follows a mechanochemical method using ball-milling. The fundamental process is established on LZC (Figures S1 and S2 exhibit the relationship between key process parameters and the impedance spectrum). A series of LZICF samples with varied InF_3 contents are synthesized based on the molar ratio $(2+x)\text{LiCl} + (1-x)\text{ZrCl}_4 + x\text{InF}_3$ ($x = 0.1, 0.25, 0.4$). For the pristine LZC ($x = 0$), its X-ray diffraction (XRD) peaks (Figure 1a) match well with the features of Li_3YCl_6 (PDF#00-044-0286), indicating the same crystal structure belonging to the $P-3m1$ space group [31]. Upon introducing InF_3 at a low doping level ($x = 0.1$), LZICF retains the same crystal structure as LZC (Figure S3), but exhibits reduced crystallinity and increased structural disorder, as evidenced by the broadened and weakened diffraction peaks (e.g., at $2\theta = 32.0^\circ$). With further increasing the doping level ($x = 0.25$ and 0.4), the features of $P-3m1$ space group disappear while two distinct peaks at 29.9° and 34.4° emerge, indicating a structural transition toward a Li_3ScCl_6 -like phase (PDF#04-009-8885) with the $C2/m$ space group [32].

This phase transition is related to the effect of the cations on the anion sublattice framework [33], which can be quantitatively described by the cation polarization factor (τ). Theoretically, τ is defined as the ratio of the summed molar content weighted ionic potentials of all cations to that of all anions. For Zr(IV)-based SSEs, τ can be expressed as:

$$\tau = \frac{5+x}{6} \frac{\phi_{\text{Li}} + \Sigma\phi_{\text{M}}}{\Sigma\phi_{\text{X}}} \quad (1)$$

where $\phi = ne/r$ represents the product of the molar content (n) of a given ion and its ionic potential (charge to radius ratio, e/r). Here, the numerator accounts for the polarizing effect of Li^+ and other cations ($\text{M} = \text{Zr}, \text{In}$), while the denominator represents the (anti-)polarizability of the anion framework ($\text{X} = \text{Cl}, \text{F}$). In the

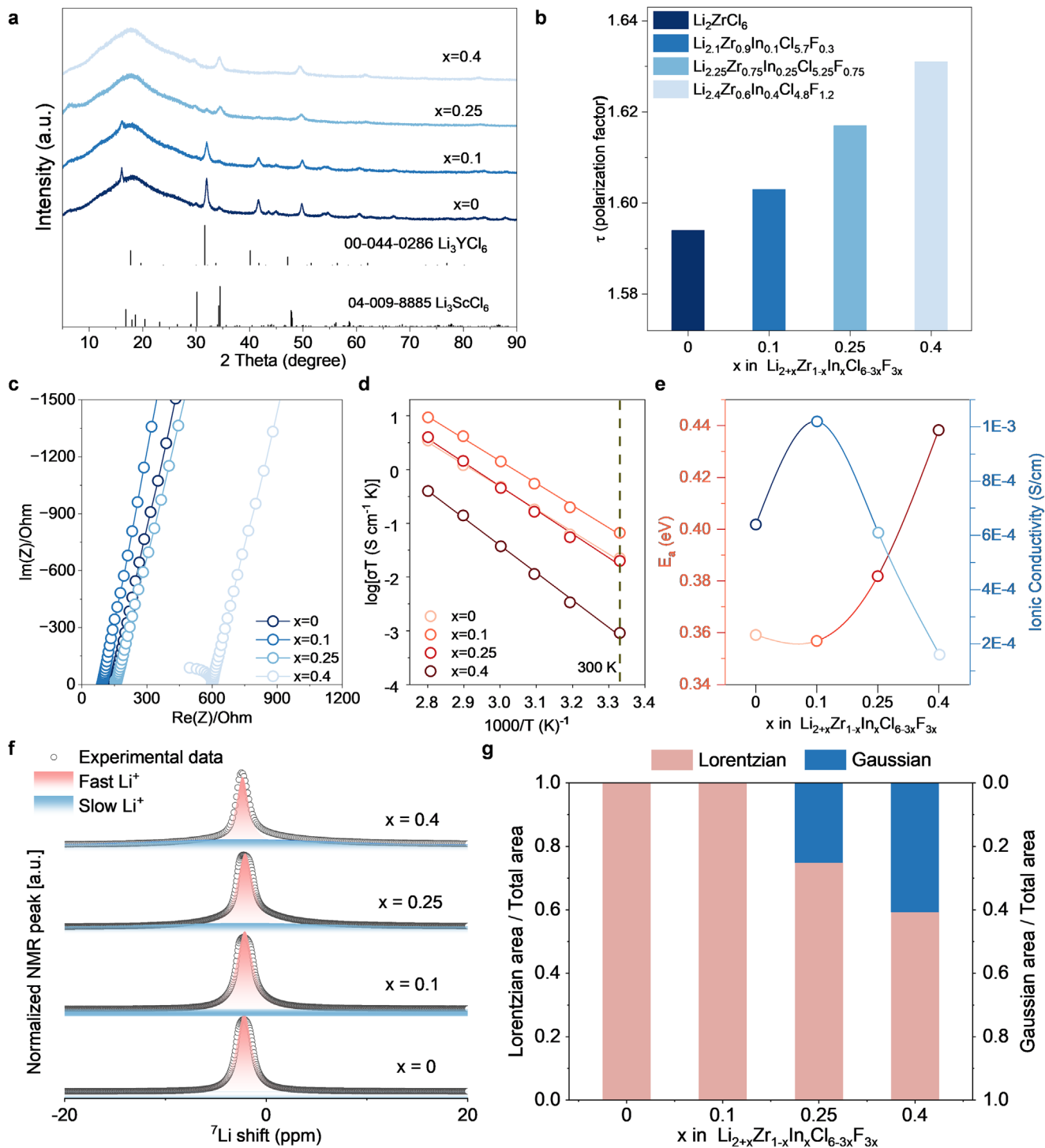


FIGURE 1 | (a) XRD patterns and (b) cation polarization factors of $\text{Li}_{2+x}\text{Zr}_{1-x}\text{In}_x\text{Cl}_{6-3x}\text{F}_{3x}$ ($x = 0, 0.1, 0.25, 0.4$). (c) Nyquist, (d) Arrhenius plots, and (e) corresponding ionic conductivity and activation energy of synthesized $\text{Li}_{2+x}\text{Zr}_{1-x}\text{In}_x\text{Cl}_{6-3x}\text{F}_{3x}$ ($x = 0, 0.1, 0.25, 0.4$) samples. (f) The solid-state NMR pattern of $\text{Li}_{2+x}\text{Zr}_{1-x}\text{In}_x\text{Cl}_{6-3x}\text{F}_{3x}$ ($x = 0, 0.1, 0.25, 0.4$). The red peak corresponds to the fast Li^+ , and the blue peak corresponds to the slow Li^+ . (g) The ratio of the Lorentzian peak (the fast Li^+) and the Gaussian peak (the slow Li^+).

specific case of $\text{Li}_{2+x}\text{Zr}_{1-x}\text{In}_x\text{Cl}_{6-3x}\text{F}_{3x}$, the value of doping content x could be directly used in the upper expression. As shown in Figure 1b, the cation polarization factors (τ) for LZC ($x = 0$) and LZICF ($x = 0.1, 0.25$, and 0.4) samples are calculated as 1.594, 1.603, 1.617, and 1.631, respectively. The monotonic increase in τ with increasing InF_3 doping content reflects a progressive weakening of cation-induced lattice polarization. According to the

τ -based predictive framework, this trend drives a structural transition from a hexagonal-close-packed anion framework ($P-3m1$ symmetry, characteristic of pristine LZC) toward a cubic-close-packed arrangement ($C2/m$ symmetry) in the co-doped samples. Consequently, the elevated τ in LZICF promotes the formation of the $C2/m$ phase with the higher symmetric cubic-close-packed anion framework.

Subsequently, the ionic conductivities of synthesized LZICF and LZC SSEs are evaluated by electrochemical impedance spectroscopy (EIS). Correspondingly, measured Nyquist plots at room temperature and Arrhenius fitting results using ionic conductivity obtained at various temperatures of all samples are presented in Figure 1c,d, respectively. As summarized in Figure 1e, the ionic conductivity and apparent activation energy (E_a) exhibit a clear dependence on the InF_3 -doping contents. Drawing on previous research experience, $C2/m$ symmetry is less favorable for ionic conduction in Zr-based halide SSE systems [17, 18]. The observed changes in ionic conductivity at higher doping levels ($x = 0.25$ and 0.4) further corroborate this conclusion. In contrast, the low doping content ($x = 0.1$) preserves the $P\text{-}3m1$ structure with reduced crystallinity exhibit the highest ionic conductivity of $1.03 \times 10^{-3} \text{ S cm}^{-1}$ and the lowest activation energy E_a of 0.357 eV at $x = 0.1$.

To further elucidate the relationship between ionic conductivity and structural characteristics, solid-state ^7Li nuclear magnetic resonance (ssNMR) spectroscopy is performed on these samples (Figure 1f). Despite differences in crystal structure, the similar ^7Li chemical shifts across all samples suggest a comparable local chemical environment for most lithium ions. However, the ^7Li NMR peak shape provides deeper insight into Li-ion dynamics. Specifically, a sharp Lorentzian peak is characteristic of highly mobile Li ions in superionic conductors, whereas a broadened Gaussian component reflects the lithium constrained in more anisotropic environments [34, 35]. The fitted Lorentzian and Gaussian contributions are shown as the red and blue components in Figure 1f, respectively. As illustrated in Figure 1g, the ssNMR spectra of LZC and low-doped LZICF ($x = 0.1$) are dominated by the Lorentzian component, indicating that the trigonal structure ($P\text{-}3m1$ space group) is well suited for fast ion transport in LZC-type halide solid electrolytes. With the increasing of doping concentration, the fraction of the Gaussian component rises significantly, reaching 25.2% for $x = 0.25$ and 40.7% for $x = 0.4$. These results confirm that the trigonal-to-monoclinic phase transition is detrimental to ionic conduction in LZC, as it introduces locally anisotropic ion transport associated with the monoclinic phase [34, 35]. In summary, low-concentration InF_3 doping ($x = 0.1$) preserves the trigonal $P\text{-}3m1$ structure while introducing structural disorder, which is believed to facilitate fast Li^+ transport and yields the highest ionic conductivity. Accordingly, the sample exhibiting the highest ionic conductivity is selected for further study and is hereafter denoted as LZICF0.1.

The local coordination environments of Zr and In in pristine LZC and LZICF0.1 are investigated using Fourier-transform extended X-ray absorption fine structure (FT-EXAFS) spectroscopy combined with wavelet transform (WT) analysis. In the Zr K-edge FT-EXAFS spectrum of pristine LZC (Figure 2a), the dominant peak located at approximately $1.8 \text{ \AA}$ is attributed to the first coordination shell of Zr–Cl bonds. Weaker features observed near 1 and $3.2 \text{ \AA}$ are consistent with multiple scattering pathways, such as Cl–Zr–Cl, characteristic of the $[\text{ZrCl}_6]^{2-}$ octahedral configuration. Upon InF_3 doping, the position of the main peak remains largely unchanged, indicating minimal variation in the Zr coordination environment. WT analysis further supports this observation, as the maximum in the WT contour plot for Zr is nearly identical before and after doping, confirming the

retention of the local coordination geometry. However, a noticeable reduction in the intensity of the peak near $1 \text{ \AA}$ is observed in the LZICF0.1 sample (Figure 2b), which suggests the decrease of scattering amplitude and local lattice deformation induced by the partial substitution of Cl^- ions ($Z = 17$, $r = 1.81 \text{ \AA}$) in the host framework by the smaller F^- ions ($Z = 9$, $r = 1.33 \text{ \AA}$). The In K-edge FT-EXAFS spectrum exhibits a prominent peak at $R \approx 2 \text{ \AA}$ ($k \approx 6 \text{ \AA}^{-1}$), corresponding to the In–Cl scattering path, as shown in Figure 2c. This confirms the successful incorporation of In^{3+} into the packed anion sublattice within the octahedral coordination environment. Due to the larger ionic radius of In^{3+} ($\sim 0.80 \text{ \AA}$) compared to Zr^{4+} ($\sim 0.72 \text{ \AA}$), its substitution induces a local expansion of the $[\text{Zr}/\text{InX}_6]$ ($X = \text{Cl}, \text{F}$) octahedra, which is accommodated by structural relaxation around the dopant sites. This local lattice distortion, originating from ionic size mismatch, mediates a reorganization of the bonds between central cations (Zr/In) and surrounding anions (Cl^-/F^-), thereby giving rise to increased structural disorder and the emergence of amorphous regions, as indicated by Fourier-transform (FT)-EXAFS spectra in Figure 2d–f.

To elucidate the underlying mechanism of ion doping modulation of Li^+ transport kinetics, ab initio molecular dynamics (AIMD) calculations are performed on LZC, single F^- -doped (LZCF), single In^{3+} -doped (LZIC), and $\text{In}^{3+}/\text{F}^-$ co-doped (LZICF0.1) systems to reveal the lattice dynamic characteristics. Figure 2g shows the vibrational density of states (vDOS) of lithium obtained by velocity autocorrelation function (VACF). Obviously, all the samples exhibit a similar vDOS distribution curve, which is due to the similar lattice framework and comparable element composition. Furthermore, the lattice vibration characteristic could be quantified by the concept of phonon band center (Figure 2h) [36, 37]. The calculated phonon band center of LZC is 237.78 cm^{-1} . Individual In^{3+} and F^- doping leads to distinct shifts in the phonon band center, moving it up to 236.67 and down to 243.64 cm^{-1} , respectively. It makes sense that In^{3+} softens the lattice due to its lower charge, whereas F^- stiffens the lattice through the stronger polarization interactions. However, the band center of dual-ion doped LZICF0.1 is located at 243.74 cm^{-1} , indicating that the stiffening effect of F^- doping plays a dominant role. This increased vibrational frequency of the lithium may be associated with modified lattice dynamics that potentially facilitate ion migration, which could contribute to the enhanced ionic conductivity of LZICF0.1.

Figure 3a displays the lithium probability isosurface in the AIMD simulation process, which is generally regarded as representing the migration trajectory in lithium superionic conductors. It is clear that although the lithium migration pathway in LZICF0.1 is similar to that in LZC, its isosurface is notably denser, implying a more active ion migration process. To understand the details of diffusion behavior, the top views of three layers with different cation arrangement characteristics denoted in Figure 3a are extracted, as presented in Figure 3b. It is evident that the intralayer percolation is smoother in LZICF0.1, which seems to suggest that the doped InF_3 accelerated the lithium diffusion in the ab -plane. Quantitatively, the diffusion coefficients are extracted from AIMD calculations and shown in Figure 3c. For LZC, the diffusion coefficients along the a -, b -, and c -axis are 1.09×10^{-5} , 8.80×10^{-6} , and $2.87 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, respectively, and the

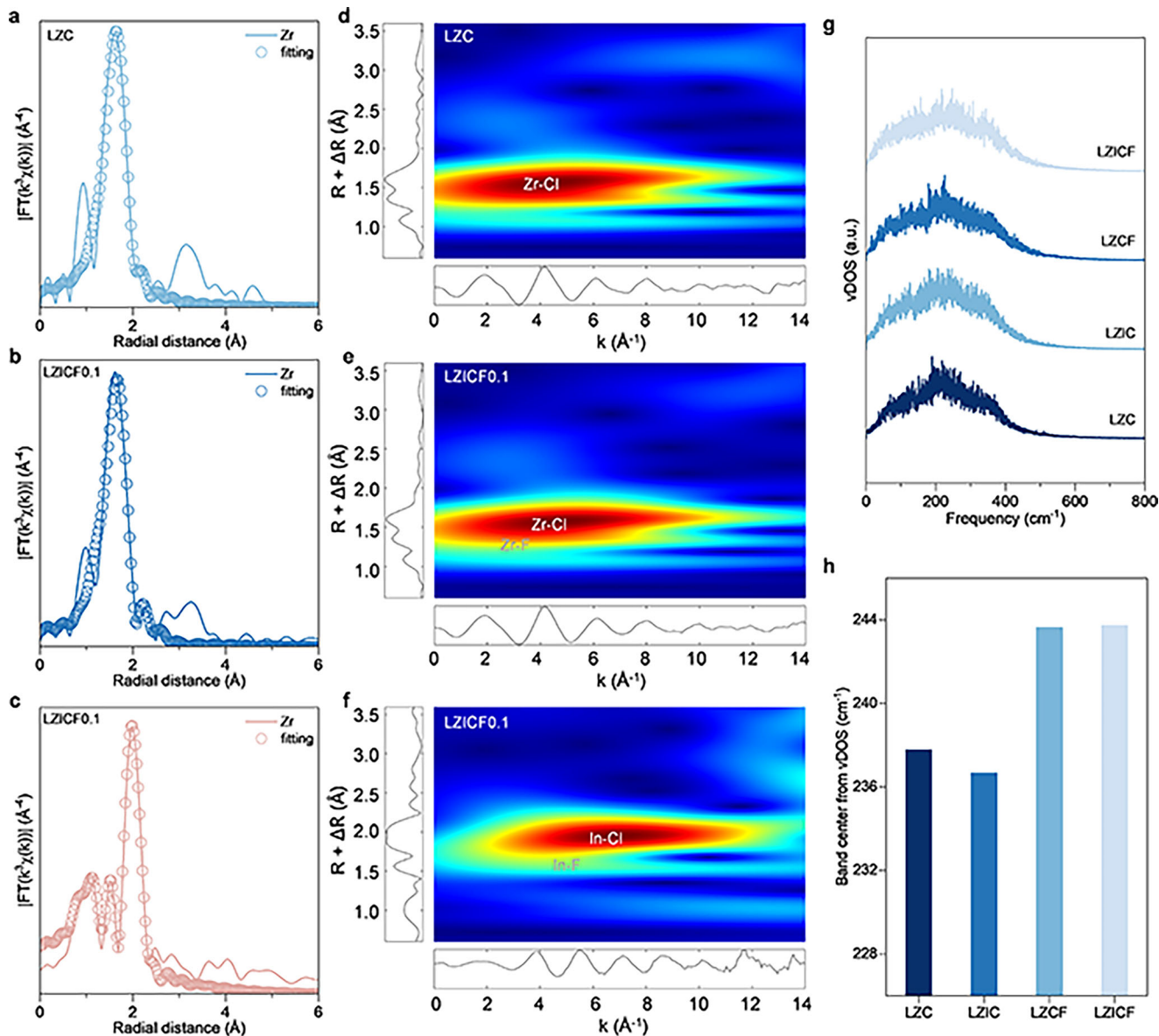


FIGURE 2 | (a–c) Wavelet-transformed EXAFS contour plots of Zr and In at Zr-edge. The original EXAFS signal $\chi(k)$ is weighted by k^2 , and k represents wavenumber. $R + \Delta R$ represents the radial distance, and ΔR indicates the distance correction due to phase shifts. (d–f) FT-EXAFS spectra of Zr and In at the Zr K-edge. (g) The vibration density of state (vDOS) of lithium ions in samples with different doping concentration. (h) The histogram of lithium-projected band center obtained from vDOS.

total bulk diffusion coefficient of LZC is $1.61 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$. It is clear that the contribution of diffusion along the c -axis accounts for a large proportion of the total diffusion in LZC, indicating the severe anisotropic characteristics of ionic migration. This phenomenon is also reported in Li_3YCl_6 , which exists within the same space group as LZC [38]. However, the diffusion coefficients of LZICF0.1 along the a -, b -, and c -axis are 1.68×10^{-5} , 2.20×10^{-5} , and $2.21 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, ultimately resulting in a higher total diffusion coefficient of $2.03 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$. This anisotropic behavior of ion transport could be depicted by the ratio of its directional diffusion coefficient to the total (Figure 3d). For LZC, $D_c/D_{\text{tot}} = 1.78$ indicates severe c -axis-dominated transport; for LZICF0.1, the ratios approach unity ($D_a/D_{\text{tot}} = 0.83$, $D_b/D_{\text{tot}} = 1.08$, $D_c/D_{\text{tot}} = 1.09$), directly demonstrating that co-doping substantially reduces anisotropy and promotes more isotropic Li^+ diffusion. Thus, in LZICF0.1, the decrease in the c -axis

diffusion coefficient is outweighed by an enhancement within the ab -plane. Consequently, the ion transport anisotropy is substantially reduced, yielding a net enhancement in lithium migration dynamics.

In addition, the Van Hove correlation function is used to statistically elucidate the physical image of lithium migration [39]. Figure 3e,f displays the self-part of the Van Hove correlation function (G_s), defined as the probability distribution of lithium-ion displacements over a specific time t [40]. The peak around $r = 1 \text{ \AA}$ represents the first-neighbor Li–Li pair, which quenches faster in LZICF0.1 than in LZC. The distinct part of the Van Hove correlation function (G_d), which describes the time-evolving radial distribution of Li ions around a central ion (Figure 3g,h). Evidently, G_d emerges and enters a relatively stable state earlier in LZICF0.1, providing evidence

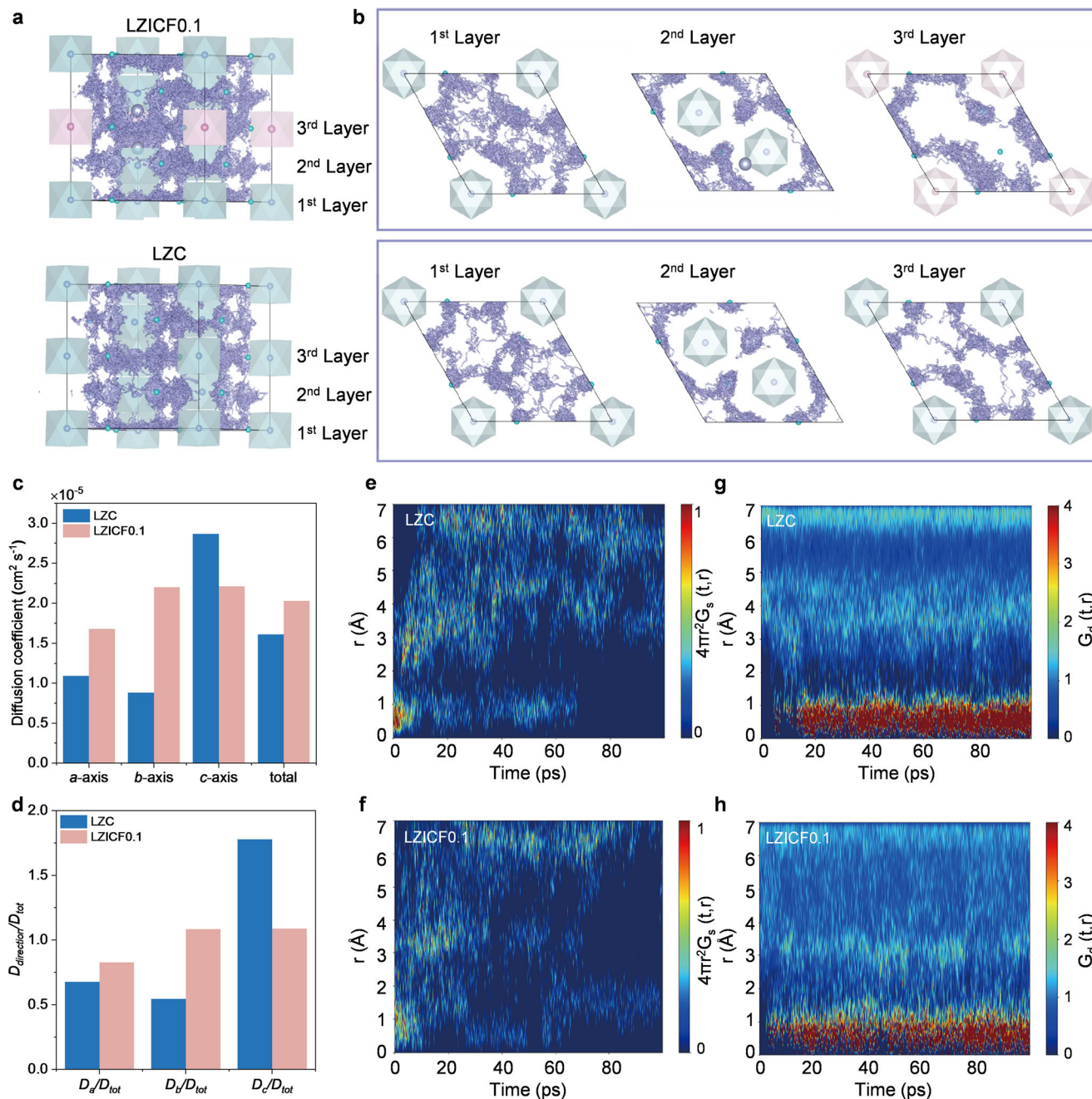


FIGURE 3 | (a) Crystal model of LZICF0.1 and LZC used in AIMD calculation and isosurface of lithium probability $P = P_{\max}/512$ at 600 K. (b) The isosurface of lithium probability in three layers with different cation arrangement characteristics. In (a) and (b), the blue and pink octahedra represent the $[\text{ZrX}_6]$ and $[\text{InX}_6]$, and the cyan and silver atoms are Li and F, respectively. The main anion framework constructed by Cl is hidden. Calculated (c) diffusion coefficients and (d) ratios of diffusion coefficients along specific directions of LZC and LZICF 0.1. The self-part of the Van Hove correlation function of (e) LZC and (f) LZICF0.1. The distinct part of the Van Hove correlation function of (g) LZC and (h) LZICF0.1.

for accelerated correlated motion of Li ions. The Van Hove correlation functions of In- or F- doped samples (denoted as LZIC and LZCF) are also calculated to study the effect of single doping (Figure S6). Although the G_d ones also show the enhanced correlated migration of Li ions, the longer delay attenuation of G_s signifies inconsistent migration behavior for the Li ions. In contrast, this phenomenon disappears in LZICF0.1, directly evidencing that the dramatic enhancement in superionic conductivity arises from a synergistic dual-ion doping effect.

The electrochemical performance of LZICF0.1 is evaluated using ASSBs employing a LiCoO_2 (LCO)-based composite cathode, a bilayer SSE of LZICF0.1 and $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSC), and a Li-In anode. As shown in Figure 4a, the LZICF0.1 cell (cathode loading 8.9 mAh cm^{-2}) exhibits discharge capacities of 137.9, 132.2, 125.7, 118.7, and 102.7 mAh g^{-1} under 0.1 C, 0.2 C, 0.5 C, 1 C, and 2 C ($1 \text{ C} = 140 \text{ mA g}^{-1}$), respectively, demonstrating markedly enhanced rate capability compared to the LZC counterpart, particularly at high current densities. The corresponding charge-discharge profiles of the LZICF0.1 cell (Figure 4b) indicate

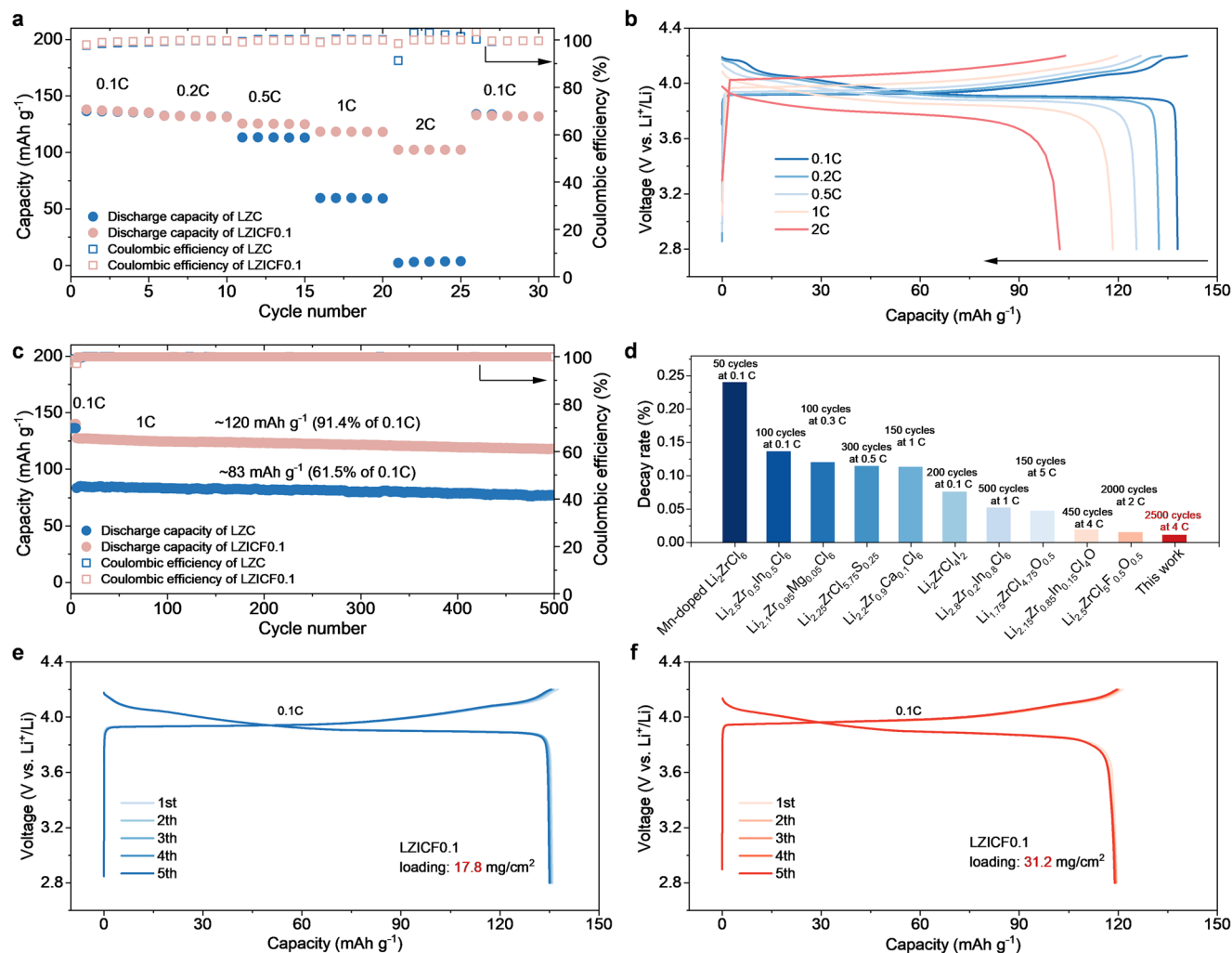


FIGURE 4 | (a) Rate performance of the LCO/LZICF0.1/LZICF0.1/LPSC/Li-In and the LCO/LZC/LZC/LPSC/Li-In cells. (b) Capacity-voltage curves of the first cycle of each rate of the LCO/LZICF0.1/LZICF0.1/LPSC/Li-In cell. (c) Long-time cycling of the LCO/LZICF0.1/LZICF0.1/LPSC/Li-In and the LCO/LZC/LZC/LPSC/Li-In cells at 1 C (5 cycles at 0.1 C are set before to activate the cells). (d) Performance comparisons with recent works on ASSBs assembled with doped-LZC SSEs. Capacity-voltage curves of the initial 5 cycles of the LCO/LZICF0.1/LZICF0.1/LPSC/Li-In cells with cathode loadings of (e) 17.8 and (f) 31.2 mg cm⁻², respectively.

good reversible behaviors and high Coulombic efficiencies at various rates. The long-cycling performance of the LZICF0.1 cell is further evaluated at 1 C (after 5 activation cycles under 0.1 C), as shown in Figure 4c. After 500 cycles, the LZICF0.1 cell maintains a discharge capacity of 117.9 mAh g⁻¹, corresponding to a 92.5% retention of the capacity delivered at the first cycle under 1 C condition. Meanwhile, the LZC cell also shows a good anti-degradation performance from 83.3 mAh g⁻¹ at the first 1 C cycle to 76.8 mAh g⁻¹ after 500 cycles. This superior performance originates from the intrinsic oxidation resistance of the anion element. Furthermore, the faster charge/discharge test is performed at 4 C. As shown in Figure S7, the LZICF0.1 cell exhibits considerable capacity retention of 72.5% with a low decay rate of 0.011% per cycle over 2500 cycles, highlighting its exceptional durability and ultrahigh rate performance. Compared with recent reports on ASSBs using doped-LZC SSEs, such high capacity retention of the LZICF0.1 cell also preserves a leading level, as shown in Figure 4d and Table S5. In addition, benefiting from its enhanced ionic conductivity, LZICF0.1 also enables the operation of thick-electrode configurations. At a cathode loading

of 17.8 mg cm⁻² (Figure 4e), the LZICF0.1 cell delivers first charge/discharge capacities of 137.53 and 135.89 mAh g⁻¹ with a high initial coulombic efficiency of 98.8%, accompanied by only a marginal increase in interfacial impedance (Figure S8). Even at a higher loading of 31.2 mg cm⁻² (Figure 4f), the cell retains competitive initial capacities of 121.29/119.57 mAh g⁻¹ and a Coulombic efficiency of 98.6%, underscoring its strong potential for high-energy-density ASSBs. Furthermore, the average areal capacities over the initial five cycles reach 2.4 and 3.7 mAh cm⁻² for electrode loadings of 17.8 and 31.2 mg cm⁻², respectively.

From an electrochemical perspective, the superior performance of the LZICF0.1 cell originates from its reduced interface resistance, as evidenced by in situ EIS and distribution of relaxation times (DRT) analysis (Figure S9). Compared to LZC, the LZICF0.1 cell exhibits suppressed grain boundary resistance and consistently lower interfacial resistance across all charge-discharge stages, confirming superior lithium transport kinetics and better interfacial compatibility within the composite cathode. Additionally, the In/F co-doping strategy also enhances the interfacial

stability against lithium metal (Figure S11). As indicated by theoretical calculations (Figure S10, Tables S2 and S3) and XPS results (Figure S13), the formation of LiF and alloy phase (e.g., Zr_3In) at the LZICF0.1/Li metal interface could suppress severe interfacial reactions and dendrite penetration. Thus, this anion-cation co-doping strategy also exhibits potential to improve reductive stability of LZC SSEs, providing a potential solution to develop ASSBs with a single halide SSE layer.

3 | Conclusion

In this work, we demonstrate that weakening ion transport anisotropy is an effective strategy to activate superionic conduction in halide SSEs. Through a rational anion-cation co-doping approach using InF_3 , the intrinsic limitation of insufficient ionic conductivity in LZC SSEs is effectively addressed. The optimized composition, LZICF0.1, achieves a high ionic conductivity of $1.03 \times 10^{-3} \text{ S cm}^{-1}$ with a low E_a of 0.357 eV. The phase transition from $P-3m1$ to $C2/m$ upon InF_3 co-doping is observed in XRD measurements. vDOS and phonon band center analysis reveal how In^{3+} and F^- differentially affect lattice stiffness and Li^+ vibrational modes, while directional diffusion coefficients and their ratios indicate that co-doping weakens the intrinsic c -axis preferential transport and activates ab -plane percolation, substantially reducing transport anisotropy. Benefiting from these merits, LZICF0.1 achieves a superior combination of high ionic conductivity, exceptional cycling stability (2500 cycles at 4 C), and high cathode loading capability (31.2 mg cm^{-2} , 3.7 mAh cm^{-2}), outperforming state-of-the-art LZC derivatives. This work not only provides a viable strategy to optimize ion transport and interfacial stability in halide SSEs, but also establishes transport anisotropy engineering as a fundamental design principle for next-generation superionic conductors.

Author Contributions

Suzhe Liang: Writing – review and editing, investigation. **Jian Peng:** writing – review and editing, investigation. **Ying Lin:** formal analysis, writing – review and editing. **Fei Liang:** methodology, data curation, writing – review and editing. **Tianrui Sun:** conceptualization, methodology, investigation, validation, writing – original draft, data curation, visualization. **Ziqing Wang:** methodology, writing – original draft, investigation, data curation, conceptualization, visualization. **Yanjie Wang:** methodology, data curation, writing – review and editing. **Jiaxuan Liao:** supervision, funding acquisition, writing – review and editing, conceptualization. **Yunzhou Wang:** methodology, writing – review and editing, data curation. **Linnan Bi:** methodology, writing – review and editing. **Xiongbang Wei:** data curation, writing – review and editing. **Changhong Wang:** conceptualization, project administration, supervision, writing – review and editing, funding acquisition. **Sizhe Wang:** writing – review and editing, supervision, conceptualization.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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